



Continuum Mechanics

Steven M. Wise

swise1@utk.edu
University of Tennessee

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Chapter 04

Kinematics

Part 3



Motions



Definition (Motion)

Let $B \subseteq \mathbb{E}^3$ be given. A smooth mapping

$$\phi : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

is called a *motion* if, for each fixed $t \in [0, \infty)$,

$$\phi_t := \phi(\cdot, t) : B \longrightarrow \mathbb{E}^3$$

is an admissible deformation and

$$\phi_0 = \phi(\cdot, 0) : B \longrightarrow B$$

is the identity map. For each $t \in [0, \infty)$, define

$$B_t := \phi_t(B) = \phi(B, t).$$

The set B_t is the deformed configuration at time t .

Configurations During a Motion



Remark

The continuity of the motion implies that, if no holes or ruptures are present in B , then no holes or ruptures appear in $\phi(B, t)$ for any $t > 0$.



The Inverse Motion

Since ϕ_t is an admissible deformation at every time, it has an inverse

$$\psi(\cdot, t) = \psi_t := \phi_t^{-1} : B_t \longrightarrow B.$$

If

$$\mathbf{x} = \phi_t(\mathbf{X}),$$

then

$$\psi_t(\mathbf{x}) = \mathbf{X}.$$

For every $t \geq 0$,

$$\mathbf{X} = \psi_t(\phi_t(\mathbf{X})) = \psi(\phi(\mathbf{X}, t), t), \quad \mathbf{X} \in B,$$

and

$$\mathbf{x} = \phi_t(\psi_t(\mathbf{x})) = \phi(\psi(\mathbf{x}, t), t), \quad \mathbf{x} \in B_t.$$



Definition (Material and Spatial Fields)

A *material field* is a function of the material point $\mathbf{X} \in B$, the initial configuration.

A *spatial field* is a function of the spatial point $\mathbf{x} \in B_t$, the deformed configuration.

Thus the arguments identify which description is being used:

$$\Omega = \Omega(\mathbf{X}, t) \quad \text{and} \quad \Gamma = \Gamma(\mathbf{x}, t).$$



Definition (Spatial Description of a Material Field)

Suppose

$$\Omega : B \times [0, \infty) \longrightarrow \mathbb{E}$$

is a material field and ϕ is a motion. The function

$$\Omega_s : B_t \times [0, \infty) \longrightarrow \mathbb{E}$$

defined by

$$\Omega_s(\mathbf{x}, t) := \Omega(\psi(\mathbf{x}, t), t), \quad \mathbf{x} \in B_t,$$

is called the *spatial description* of the material field Ω .



Definition (Material Description of a Spatial Field)

Suppose

$$\Gamma : B_t \times [0, \infty) \longrightarrow \mathbb{E}$$

is a spatial field. The function

$$\Gamma_m : B \times [0, \infty) \longrightarrow \mathbb{E}$$

defined by

$$\Gamma_m(\mathbf{X}, t) := \Gamma(\phi(\mathbf{X}, t), t), \quad \mathbf{X} \in B,$$

is called the *material description* of the spatial field Γ .



Material and Spatial Derivatives

We must distinguish derivatives with respect to material coordinates

$$\mathbf{X} = (X_1, X_2, X_3)^T$$

from derivatives with respect to spatial coordinates

$$\mathbf{x} = (x_1, x_2, x_3)^T.$$

We use

$$\nabla^{\mathbf{X}} \quad \text{and} \quad \nabla^{\mathbf{x}}$$

to denote derivatives with respect to material and spatial coordinates, respectively.

The same care is required when differentiating with respect to time: one must specify whether the material point $\mathbf{X}$ or the spatial point $\mathbf{x}$ is held fixed.



Time Derivatives



Definition (Total Time Derivative)

The *total time derivative* of a field is the rate of change of that field measured by an observer who follows the motion of each particle in the body.

If

$$\Omega : B \times [0, \infty) \longrightarrow \mathbb{E}$$

is a material field, then its total time derivative is

$$\dot{\Omega}(\mathbf{X}, t) := \frac{\partial}{\partial t} \Omega(\mathbf{X}, t),$$

where the material point $\mathbf{X}$ is held fixed.



Definition (Total Time Derivative of a Spatial Field)

Let $\Gamma : B_t \times [0, \infty) \rightarrow \mathcal{E}$ be a spatial field, and let Γ_m be its material description. Its total time derivative, still denoted by a superposed dot, is

$$\dot{\Gamma}(\mathbf{x}, t) := [\dot{\Gamma}_m]_s(\mathbf{x}, t). \quad (4.16)$$

Thus we first convert the spatial field to its material description, differentiate with respect to time while holding $\mathbf{X}$ fixed, and then convert the result back to a spatial description.

In general,

$$\frac{\partial}{\partial t} \Gamma(\mathbf{x}, t) \neq \dot{\Gamma}(\mathbf{x}, t).$$



Definition (Material Velocity and Acceleration)

Let $\phi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion. The vector

$$\mathbf{V}(\mathbf{X}, t) := \dot{\phi}(\mathbf{X}, t) = \frac{\partial}{\partial t} \phi(\mathbf{X}, t)$$

is the velocity at time t of the material particle labeled by $\mathbf{X} \in B$.

The vector

$$\mathbf{A}(\mathbf{X}, t) = \dot{\mathbf{V}}(\mathbf{X}, t) = \frac{\partial}{\partial t} \mathbf{V}(\mathbf{X}, t)$$

is the acceleration of that particle.



Definition (Spatial Velocity and Acceleration)

The spatial descriptions of $\mathbf{V}$ and $\mathbf{A}$ are the *spatial velocity field* and *spatial acceleration field*:

$$\begin{aligned}\mathbf{v}(\mathbf{x}, t) &:= \mathbf{V}_s(\mathbf{x}, t), & \mathbf{x} \in B_t, & \quad t \geq 0, \\ \mathbf{a}(\mathbf{x}, t) &:= \mathbf{A}_s(\mathbf{x}, t), & \mathbf{x} \in B_t, & \quad t \geq 0.\end{aligned}\tag{4.17}$$

Equivalently,

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{V}(\boldsymbol{\psi}(\mathbf{x}, t), t), \quad \mathbf{a}(\mathbf{x}, t) = \mathbf{A}(\boldsymbol{\psi}(\mathbf{x}, t), t).$$



Result 4.7 (Total Time Derivative of Spatial Fields)

Let $\phi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity $\mathbf{v}$. Suppose

$$q : B_t \times [0, \infty) \rightarrow \mathbb{R}, \quad \mathbf{w} : B_t \times [0, \infty) \rightarrow \mathcal{V}$$

are spatial fields. Then

$$\dot{q} = \frac{\partial q}{\partial t} + \nabla^x q \cdot \mathbf{v}$$

and

$$\dot{\mathbf{w}} = \frac{\partial \mathbf{w}}{\partial t} + (\nabla^x \mathbf{w})\mathbf{v}.$$



Proof.

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. Then

$$\nabla^x q = \frac{\partial q}{\partial x_i} \mathbf{e}_i, \quad \nabla^x \mathbf{w} = \frac{\partial w_i}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j.$$

By definition,

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{V}_s(\mathbf{x}, t).$$

Recall also that

$$\mathbf{V}(\mathbf{X}, t) = \frac{\partial}{\partial t} \boldsymbol{\phi}(\mathbf{X}, t),$$

so that

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{V}(\boldsymbol{\psi}(\mathbf{x}, t), t).$$



Proof (Cont.)

The material description of q is

$$q_m(\mathbf{X}, t) = q(\boldsymbol{\phi}(\mathbf{X}, t), t).$$

By the chain rule,

$$\begin{aligned}\dot{q}_m(\mathbf{X}, t) &= \frac{\partial}{\partial t} q(\boldsymbol{\phi}(\mathbf{X}, t), t) \\ &= \frac{\partial q}{\partial x_i}(\boldsymbol{\phi}(\mathbf{X}, t), t) \frac{\partial \phi_i}{\partial t}(\mathbf{X}, t) + \frac{\partial q}{\partial t}(\boldsymbol{\phi}(\mathbf{X}, t), t) \\ &= \frac{\partial q}{\partial x_i}(\boldsymbol{\phi}(\mathbf{X}, t), t) V_i(\mathbf{X}, t) + \frac{\partial q}{\partial t}(\boldsymbol{\phi}(\mathbf{X}, t), t).\end{aligned}$$

Here i is a dummy index.



Proof (Cont.)

Returning to the spatial description gives

$$\begin{aligned}
 \dot{q}(\mathbf{x}, t) &= [\dot{q}_m]_s(\mathbf{x}, t) \\
 &= [\dot{q}_m(\mathbf{X}, t)]_{\mathbf{X}=\boldsymbol{\psi}(\mathbf{x}, t)} \\
 &= \frac{\partial q}{\partial x_i}(\mathbf{x}, t) v_i(\mathbf{x}, t) + \frac{\partial q}{\partial t}(\mathbf{x}, t) \\
 &= \nabla^x q(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) + \frac{\partial q}{\partial t}(\mathbf{x}, t).
 \end{aligned}$$

The calculation for $\dot{\mathbf{w}}$ is componentwise identical and gives

$$\dot{\mathbf{w}} = \frac{\partial \mathbf{w}}{\partial t} + (\nabla^x \mathbf{w}) \mathbf{v}.$$





Remark (Partial-Derivative Notation)

For a spatial field $q(\mathbf{x}, t)$, the notation

$$\frac{\partial q}{\partial t}$$

means the derivative with respect to the variable in the second position, with $\mathbf{x}$ fixed.

Likewise,

$$\frac{\partial q}{\partial x_i}$$

means the derivative with respect to the i th coordinate of the variable in the first position, with t fixed.



Corollary (Spatial Acceleration)

The spatial acceleration field satisfies

$$\mathbf{a}(\mathbf{x}, t) = \dot{\mathbf{v}}(\mathbf{x}, t) = \frac{\partial \mathbf{v}}{\partial t}(\mathbf{x}, t) + (\nabla^x \mathbf{v}(\mathbf{x}, t))\mathbf{v}(\mathbf{x}, t).$$

Proof.

An easy exercise: apply Result 4.7 to the spatial velocity field $\mathbf{v}$ and use $\mathbf{a} = \dot{\mathbf{v}}$. □

The two terms

$$\frac{\partial \mathbf{v}}{\partial t} + (\nabla^x \mathbf{v})\mathbf{v}$$

will be fundamental in the Navier–Stokes equations.



Rates of Strain and Rotation



Definition (Rate of Strain)

Let $\phi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity field $\mathbf{v}$. The *rate-of-strain tensor* is

$$L(\mathbf{x}, t) := \text{sym}(\nabla^{\mathbf{x}} \mathbf{v}(\mathbf{x}, t)) = \frac{1}{2} \left(\nabla^{\mathbf{x}} \mathbf{v} + (\nabla^{\mathbf{x}} \mathbf{v})^T \right) (\mathbf{x}, t).$$

The tensor L is symmetric.



Definition (Rate of Rotation)

The *rate-of-rotation tensor*, or *spin tensor*, is

$$W(\mathbf{x}, t) := \text{skw}(\nabla^x \mathbf{v}(\mathbf{x}, t)) = \frac{1}{2} \left(\nabla^x \mathbf{v} - (\nabla^x \mathbf{v})^T \right) (\mathbf{x}, t).$$

The tensor W is skew-symmetric, and the spatial velocity gradient has the decomposition

$$\nabla^x \mathbf{v} = L + W.$$



Example (Exercise 30)

Suppose $\mathbf{x} = \boldsymbol{\phi}(\mathbf{X}, t)$ is defined by

$$x_1 = \cos(\omega t)X_1 + \sin(\omega t)X_2,$$

$$x_2 = -\sin(\omega t)X_1 + \cos(\omega t)X_2,$$

$$x_3 = (1 + \alpha t)X_3.$$

Find

- (a) the inverse motion $\boldsymbol{\psi}$,
- (b) the spatial velocity $\mathbf{v}$,
- (c) the rate-of-strain tensor L and spin tensor W .



Example (Exercise 30(a))

The motion can be written

$$\mathbf{x} = F(t)\mathbf{X},$$

where

$$F(t) = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) & 0 \\ -\sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 + \alpha t \end{bmatrix}.$$

Provided $1 + \alpha t \neq 0$,

$$F^{-1}(t) = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & \frac{1}{1 + \alpha t} \end{bmatrix}.$$



Example (Exercise 30(a), Cont.)

Since

$$\mathbf{X} = F^{-1}(t)\mathbf{x},$$

the inverse motion is

$$\boldsymbol{\psi}(\mathbf{x}, t) = F^{-1}(t)\mathbf{x}.$$

Thus,

$$X_1 = \cos(\omega t)x_1 - \sin(\omega t)x_2,$$

$$X_2 = \sin(\omega t)x_1 + \cos(\omega t)x_2,$$

$$X_3 = \frac{x_3}{1 + \alpha t}.$$



Example (Exercise 30(b))

The material velocity is

$$\begin{aligned}\mathbf{V}(\mathbf{X}, t) &= \frac{\partial}{\partial t}(F(t)\mathbf{X}) \\ &= F'(t)\mathbf{X}.\end{aligned}$$

By definition, the spatial velocity is

$$\begin{aligned}\mathbf{v}(\mathbf{x}, t) &= \mathbf{V}(\boldsymbol{\psi}(\mathbf{x}, t), t) \\ &= F'(t)F^{-1}(t)\mathbf{x}.\end{aligned}$$



Example (Exercise 30(b), Cont.)

Direct multiplication gives

$$F'(t)F^{-1}(t) = \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & \frac{\alpha}{1+\alpha t} \end{bmatrix}.$$

Therefore,

$$\mathbf{v}(\mathbf{x}, t) = \omega x_2 \mathbf{e}_1 - \omega x_1 \mathbf{e}_2 + \frac{\alpha x_3}{1+\alpha t} \mathbf{e}_3.$$

The components are extracted using

$$v_i = [\mathbf{v}]_i.$$



Example (Exercise 30(c))

From the spatial velocity,

$$\nabla^x \mathbf{v} = \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & \frac{\alpha}{1+\alpha t} \end{bmatrix}.$$

Consequently,

$$L = \text{sym}(\nabla^x \mathbf{v}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{\alpha}{1+\alpha t} \end{bmatrix}$$

and

$$W = \text{skw}(\nabla^x \mathbf{v}) = \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$



Example (Exercise 30(c), Cont.)

The rate of strain depends only on the stretching in the $\mathbf{e}_3$ direction:

$$L\mathbf{e}_3 = \frac{\alpha}{1 + \alpha t} \mathbf{e}_3, \quad L\mathbf{e}_1 = L\mathbf{e}_2 = \mathbf{0}.$$

The spin tensor depends only on the rotation rate ω :

$$W\mathbf{e}_1 = -\omega\mathbf{e}_2, \quad W\mathbf{e}_2 = \omega\mathbf{e}_1, \quad W\mathbf{e}_3 = \mathbf{0}.$$

Remark

See Exercise 31 for further interpretive details concerning L and W .



Definition (Vorticity)

Let $\phi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity field $\mathbf{v}$. The *vorticity* of the motion is

$$\mathbf{w} := \nabla^x \times \mathbf{v}.$$

The curl uses its linearity and its action on a right-handed orthonormal frame. In components,

$$[\mathbf{w}]_j = \epsilon_{ijk} \frac{\partial v_i}{\partial x_k},$$

where j is free and i, k are dummy indices.



Proposition

Let $\phi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity $\mathbf{v}$. The vorticity $\mathbf{w}$ is the axial vector of the skew-symmetric tensor $2W$, where W is the spin tensor.

Proof.

By definition,

$$W = \text{skw}(\nabla^x \mathbf{v}),$$

so

$$2W = \nabla^x \mathbf{v} - (\nabla^x \mathbf{v})^T.$$

The defining property of the curl gives, for every $\mathbf{a} \in \mathcal{V}$,

$$\begin{aligned} (\nabla^x \times \mathbf{v}) \times \mathbf{a} &= \left(\nabla^x \mathbf{v} - (\nabla^x \mathbf{v})^T \right) \mathbf{a} \\ &= (2W)\mathbf{a}. \end{aligned}$$

By uniqueness of the axial vector, $\mathbf{w}$ is the axial vector of $2W$. □



Rigid Motion



Definition (Rigid Motion)

A motion

$$\phi : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

is called a *rigid motion* if

$$\phi(\mathbf{X}, t) = \mathbf{c}(t) + R(t)\mathbf{X},$$

where, for every $t \geq 0$,

$$R(t)R^T(t) = I, \quad \det R(t) = 1.$$

Thus $\mathbf{c}(t)$ gives a translation and $R(t)$ gives a proper rotation.



Proposition

Suppose $\phi : B \times [0, \infty) \rightarrow \mathbb{E}^3$ is a rigid motion and $\mathbf{v}$ is its spatial velocity field. Then

$$\mathbf{v}(\mathbf{x}, t) = \boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t),$$

where $\boldsymbol{\omega}(t)$ is a time-dependent vector called the spatial angular velocity.

Equivalently,

$$\mathbf{v}(\mathbf{x}, t) = \Omega(t)(\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t), \quad (4.18)$$

where

$$\Omega(t) = \dot{R}(t)R^T(t)$$

is the unique skew-symmetric second-order tensor whose axial vector is $\boldsymbol{\omega}(t)$.



Proposition (Rigid Motion, Cont.)

For a rigid motion,

$$\nabla^x \mathbf{v}(\mathbf{x}, t) = \Omega(t).$$

Therefore,

$$L = O, \quad W = \Omega(t),$$

and

$$\mathbf{w} = \nabla^x \times \mathbf{v} = 2\boldsymbol{\omega}(t).$$

On the left, $\mathbf{w}$ denotes the vorticity field; on the right, $\boldsymbol{\omega}(t)$ denotes the spatial angular velocity.



Proof.

Since ϕ is a rigid motion,

$$\phi(\mathbf{X}, t) = \mathbf{c}(t) + R(t)\mathbf{X}.$$

Hence its material velocity is

$$\mathbf{V}(\mathbf{X}, t) = \frac{\partial}{\partial t} \phi(\mathbf{X}, t) = \dot{\mathbf{c}}(t) + \dot{R}(t)\mathbf{X}.$$

The inverse motion is

$$\psi(\mathbf{x}, t) = R^T(t)(\mathbf{x} - \mathbf{c}(t)),$$

because

$$\begin{aligned} \phi(\psi(\mathbf{x}, t), t) &= \mathbf{c}(t) + R(t)R^T(t)(\mathbf{x} - \mathbf{c}(t)) \\ &= \mathbf{x}. \end{aligned}$$



Proof (Cont.)

Therefore, the spatial velocity is

$$\begin{aligned}\mathbf{v}(\mathbf{x}, t) &= \mathbf{V}(\boldsymbol{\psi}(\mathbf{x}, t), t) \\ &= \dot{\mathbf{c}}(t) + \dot{R}(t)R^T(t)(\mathbf{x} - \mathbf{c}(t)).\end{aligned}$$

Define

$$\Omega(t) := \dot{R}(t)R^T(t).$$

It follows that

$$\mathbf{v}(\mathbf{x}, t) = \Omega(t)(\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t),$$

which is equation (4.18).

Differentiating $R(t)R^T(t) = I$ gives

$$\dot{R}(t)R^T(t) + R(t)\dot{R}^T(t) = O.$$



Proof (Cont.)

Since

$$\Omega^T(t) = R(t)\dot{R}^T(t),$$

the preceding identity gives

$$\Omega^T(t) = -\dot{R}(t)R^T(t) = -\Omega(t).$$

Thus $\Omega(t)$ is skew-symmetric. By the axial-vector result, there is a unique vector $\boldsymbol{\omega}(t)$ such that

$$\Omega(t)\mathbf{a} = \boldsymbol{\omega}(t) \times \mathbf{a} \quad \text{for every } \mathbf{a} \in \mathcal{V}.$$

Taking $\mathbf{a} = \mathbf{x} - \mathbf{c}(t)$ in equation (4.18) yields

$$\mathbf{v}(\mathbf{x}, t) = \boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t).$$

Hence $\boldsymbol{\omega}(t)$ is the spatial angular velocity.



Proof (Cont.)

In components, equation (4.18) is

$$v_i(\mathbf{x}, t) = \Omega_{ij}(t)(x_j - c_j(t)) + \dot{c}_i(t),$$

where i is free and j is a dummy index. Therefore,

$$\begin{aligned} \frac{\partial v_i}{\partial x_k}(\mathbf{x}, t) &= \Omega_{ij}(t) \frac{\partial}{\partial x_k} (x_j - c_j(t)) \\ &= \Omega_{ij}(t) \delta_{jk} \\ &= \Omega_{ik}(t). \end{aligned}$$

Here i, k are free in the final expression and j is a dummy index. Consequently,

$$\nabla^x \mathbf{v}(\mathbf{x}, t) = \Omega(t).$$



Proof (Cont.)

Since $\Omega^T = -\Omega$,

$$L = \text{sym}(\nabla^x \mathbf{v}) = \frac{1}{2}(\Omega + \Omega^T) = O,$$

$$W = \text{skw}(\nabla^x \mathbf{v}) = \frac{1}{2}(\Omega - \Omega^T) = \Omega.$$

The preceding proposition states that the vorticity $\mathbf{w}$ is the axial vector of $2W$. For every $\mathbf{a} \in \mathcal{V}$,

$$\begin{aligned} (2W)\mathbf{a} &= 2\Omega\mathbf{a} \\ &= 2(\boldsymbol{\omega}(t) \times \mathbf{a}) \\ &= (2\boldsymbol{\omega}(t)) \times \mathbf{a}. \end{aligned}$$

By uniqueness of the axial vector,

$$\mathbf{w} = 2\boldsymbol{\omega}(t).$$

